



2108 - Constraining the Seeding and Growth of First Black Holes via Observable Signatures from the Early Universe

Cycle: 1, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

The physical processes governing the formation of the seeds of supermassive black holes, as well as how they grow in the early universe, are two fundamental unanswered questions in galaxy formation and cosmology. The unprecedented sensitivity of JWST's instruments hold exciting potential for probing the growth of galaxies and black holes (BHs) in the very early Universe. However, detailed theoretical models are essential for interpreting these observations. With the goal of establishing a connection between the "ground-level", small-scale physical processes and the "top-level" observable signatures, we propose to construct an efficient, physics-based modeling pipeline that self-consistently simulates the co-evolution

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of BHs and galaxies, and produces predictions of their physical and observable properties over a wide redshift and halo mass range. This will be done by incorporating two essential new components into a well-established galaxy formation framework: 1) a suite of models representing different scenarios for BH seed formation and BH accretion, and 2) population synthesis and nebular line emission models that account for radiation from both stars and accreting BH. With this novel modeling pipeline, we will be able to 1) explore the implications of different BH seeding and accretion models for physical BH and host properties at high redshift, as well as observable quantities such as emission line luminosity functions, colors, and line ratios; 2) identify observables that can optimally constrain seeding and accretion models; and 3) create mock catalogs containing synthetic spectra and photometry, which can guide the design of future JWST observational programs.

OBSERVING DESCRIPTION

N/A